Al-Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Dept.

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Microprocessor Systems
3 Credit Hours
13 Dec. 2021

Mid Exam
30 Point

18

Question #1: First Question (14 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
C	A	A	B	B	B	A	A	A		A	D	C	A

1. Data bus width of Intel 8086 microprocessor is:
   A. 16 B. 24 C. 20 D. 32
2. One of the following control transfer instruction is not used with signed number
   A. JGE B. JBE C. JL D. JLE
3. The segment that contains the assembly language instructions to be executed by the microprocessor is
   A. Data segment B. Code segment C. Stack segment D. Extra segment
4. Instruction Pointer (IP) contains offset address of _____ segment.
   A. Data segment B. Code segment C. Stack segment D. Extra segment
5. If the offset address of X in data segment is equal to 105H, and DS=1000H, then the physical address of X =?
   A. 10104 H B. 10105 H C. 1105 H D. 11015H
6. If offset is a 10-bit number address, then the maximum possible locations are_____.
   A. 1024 KB B. 1024 bytes C. 64 KB D. 1 MB
7. If the size of memory is 64K, what will be the starting and ending offset addresses of it?
   A. 0000H to FFFFH B. 0000H to AFFFH C. 00000H to FFFFFH D. 000H to FFFH
8. The content of register BL after execution the instruction MOV BL,-11 :
   A. BL=F5H B. BL=FAH C. BL=F7H D. BL=FCH
9. In the following instruction sequence, show the value of AL, in binary:

```
MOV AL,3FH
NEG AL
XOR AL,FFH
```

   A. AL= 00111110B B. AL= 01011111B
   C AL= 01011110B D. AL= 01011111B
10. In the following instruction sequence, show the values of the Carry, and Zero flags :

```
MOV AL,00110011b
TEST AL,8
```

CF=0 ZF=0

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11. Find the content of AL and zero flag (ZF) after execution of the following code:

```
MOV AL,FDH
ADD AL,3
ADC AL,3
```

A. AL=4 ; ZF=0 B.AL=2 ; ZF=1 C. AL=5; ZF=0 D.AL=6 ; ZF=1

12. The equivalent code of NOT AX instruction is:

A. ADD AX,1 B.AND AX,FFFFH C. OR AX,0000H D. XOR AX,FFFFH

13. If the SP register contains the hexadecimal value F104H, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX,BX
POP AX
POP BX
```

A. F100H B. 102H C. F104H D. F106H

14. What will be the final value in AX and DX after this code executes?

```
    MOV DX,1
    MOV AL,-5
    CMP AL,000H
    JB L1
    MOV DX,0
L1: MOV AX,0
```

A. AX=0; DX=0 B. AX=0; DX=1 C. AX=-5; DX=0 D. AX=1; DX=1

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Question #2:

A) Defined below X memory locations and the offset address of X is 100h. Find the content of register AL and DX after execution of the following code?

```
X  DW 35FAH, 48CBH, 231AH, 0080H
   MOV BX, OFFSET X
   ADD BX,2
   MOV AL,[BX]
   INC BX
   MOV DX,[BX]
```

AL=231AH ; DX= 0080H

B) Defined below X memory locations and the offset address of X is 100h. Find the content of register AL and BL after execution of the following code?

```
X  DB 88H,44H,50H,90H,55H,23H,51H
   MOV BX,OFFSET X
   MOV SI,0
   MOV AL, [BX+4]
   MOV CL,X[SI]
```

AL= 8CH CL= 88H

C)
```
   MOV CL,03H
   MOV AL,127
   ROR AL,01H
   JNC L
   ADC CL,00H
L: SHR AL,CL
```

AX= 00F7H ; CL=03H ; CF= 1 ZF= 0 PF= 0

D)
```
   MOV AL,14
   MOV AH,0
   MOV BL,17
   DIV BL
```

AL= 0 H AH= E H

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Question #3

Suppose we have a microprocessor with 21 bits address bus:

a) What is the maximum memory capacity will be?

$2^n =$ 4

b) For the following instruction MOV [BX], AL; what is the addressing mode used?

- Base segment addressing mode.

c) Calculate the physical address of memory location assuming that DS=8C00h and BX= 4000h

Physical = logical * 10H + offset

4000h = 2000 * 10H + offset offset = 2

Physical = 2000 * 10H + 2

= 20002

e) Write the equivalent code of LOOP L instruction is

1.5

physical = logical * 10H + offset

20000 = 4000 * 10H + offset

offset = 5

Physical = 4000 * 10H + 5

= 50005

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